INCH-POUND
MIL-STD-1894B
26 June 1998
SUPERSEDING
MIL-STD-1894A
17 October 1986

DEPARTMENT OF DEFENSE STANDARD PRACTICE

RADIOGRAPHIC REFERENCE STANDARDS AND RADIOGRAPHIC PROCEDURES FOR PARTIAL-PENETRATION STEEL WELDS

AMSC N/A AREA NDTI

DISTRIBUTION STATEMENT A. Approved for public release; distribution is unlimited.

Source: https://assist.dla.mil -- Downloaded: 2026-10-26T07:55Z
Check the source to verify that this is the current version before use.

FOREWORD

1. This standard is approved for use by all Departments and Agencies of the Department of Defense.

2. This document contains selected radiographs illustrating various types and degrees of discontinuities occurring in partial-penetration steel welds. It also contains instructional data concerning radiographic inspection of this type of weldment.

3. It is intended that this document provide the following:

   a. A weld discontinuity severity range for designers to select standards for acceptance inspection.
   b. Reference standards for acceptance personnel to evaluate the quality of production weldments.
   c. Guide lines for radiographers to effectively accomplish radiographic examination of partial-penetration welds.

4. Beneficial comments (recommendations, additions, deletions) and any pertinent data which may be of use in improving this document should be addressed to: U.S. Army Tank-automotive and Armaments Command, ATTN: AMSTA-TR-E/BLUE, Warren, MI 48397-5000, by using the Standardization Document Improvement Proposal (DD Form 1426) appearing at the end of this document, or by letter.

Source: https://assist.dla.mil -- Downloaded: 2026-10-26T07:55Z
Check the source to verify that this is the current version before use.

CONTENTS

PARAGRAPH		PAGE
	FOREWORD	ii
1.	SCOPE	1
1.1	Scope	1
1.2	Use of radiographs	1
2.	APPLICABLE DOCUMENTS	2
2.1	General	2
2.2	Government documents	2
2.3	Non-Government publications	2
2.4	Order of precedence	3
3.	DEFINITIONS	4
3.1	Metallographic cross - sections	4
3.2	Partial penetration joint design	4
3.3	Penetrameter / Image quality indicator	4
3.4	Nondestructive testing terms and definitions.	4
4.	GENERAL REQUIREMENTS	5
4.1	Radiographic location	5
4.1.1	Radiographic frequency	5
4.1.2	Angle beam	5
4.1.3	Penetrameter requirements	5
4.2	Acceptance standards	5
4.3	Discontinuity types	5
4.4	Discontinuity grading	6
5.	DETAILED REQUIREMENTS	7
5.1	Application of reference standards	7
5.1.1	Determination of acceptability	7
5.1.2	Quality levels of radiographic sensitivity	7
5.1.3	Rules 1, 2, and 3	7
5.1.4	False indications	8
5.2	Inspection records	8
6.	NOTES	9
6.1	Intended use	9
6.2	Subject (key word) listing	9
6.3	Changes from previous issue	9

Source: https://assist.dla.mil -- Downloaded: 2026-10-26T07:55Z
Check the source to verify that this is the current version before use.

FIGURES

FIGURE(S)		PAGE(S)
1	Correct and incorrect radiographic procedure for joint design (A)	10
2	Reference standards for fine scattered porosity in steel welds	11-12
3	Reference standards for coarse scattered porosity in steel welds	13-14
4	Reference standards for linear porosity in steel welds	15-16
5	Reference standards for clustered porosity in steel welds	17
6	Reference standards for scattered slag inclusions in steel welds	18-19
7	Varying degrees of incomplete penetration in steel welds correlated with weld cross-sections.	20
8	Lack of fusion in steel welds correlated with weld cross-sections	21
9	Recommended angle beam and film location for typical partial-penetration joints	22-24

Source: https://assist.dla.mil -- Downloaded: 2026-10-26T07:55Z
Check the source to verify that this is the current version before use.

1. SCOPE

1.1 Scope. This standard provides standard reference radiographs and recommended radiographic inspection procedures for partial-penetration weldments in steel plate, casting, or forging, 0.5 to 2 inches (in.) (12.70 to 50.80 millimeters (mm)) thick.

1.2 Use of radiographs. The reference radiographs comprised in this standard are intended primarily for use on manual or machine arc welding with the gas metal arc, gas tungsten arc, and submerged arc welding processes.

Source: https://assist.dla.mil -- Downloaded: 2026-10-26T07:55Z
Check the source to verify that this is the current version before use.

2. APPLICABLE DOCUMENTS

2.1 General. The documents listed below are not necessarily all of the documents referenced herein, but are the ones that are needed in order to fully understand the information provided by this standard.

2.2 Government documents.

2.2.1 Specifications, standards, and handbooks. The following specifications, standards, and handbooks form a part of this document to the extent specified herein. Unless otherwise specified, the issues of the documents which are DoD adopted are those listed in latest issue of the Department of Defense Index of Specifications and Standards (DoDISS) and supplements thereto.

STANDARDS

DEPARTMENT OF DEFENSE

MIL-STD-410	- Nondestructive Testing Personnel Qualification and Certification.
MIL-STD-1941	- Metal Arc Welding of Homogenous Armor.

(Note: Negatives of radiographs illustrated in this standard can be purchased through AMSTA-TR-E/BLUE. Requests should be addressed to U.S. Army Tank-automotive and Armaments Command, Attn: AMSTA-TR-/BLUE, Warren, MI 48397-5000.)

(Unless otherwise indicated, copies of the above specifications, standards, and handbooks are available from the Standardization Document Order Desk, 700 Robbins Avenue, Building 4D, Philadelphia, PA 19111-5094.)

2.3 Non-Government publications. The following documents form a part of this document to the extent specified herein. Unless otherwise specified, the issues of the documents which are DoD adopted are those listed in latest issue of the Department of Defense Index of Specifications and Standards (DoDISS) and supplements thereto.

AMERICAN SOCIETY FOR TESTING AND MATERIALS (ASTM)

ASTM E142	- Controlling Quality of Radiographic Testing, Method for (DoD Adopted).
ASTM E340	- Standard Method for Macroetching Metals and Alloys (DoD Adopted).

Source: https://assist.dla.mil -- Downloaded: 2026-10-26T07:55Z
Check the source to verify that this is the current version before use.

ASTM E1316	- Standard Terminology for Nondestructive Examinations.
ASTM E1742	- Radiographic Examination, Standard Practice for (DoD Adopted).

(Application for copies should be addressed to the American Society for Testing and Materials, 100 Barr Harbor Drive, West Conshohocken, PA 19428-2959.)

2.4 Order of precedence. In the event of a conflict between the text of this document and the references cited herein, the text of this document takes precedence. Nothing in this document, however, supersedes applicable laws and regulations unless a specific exemption has been obtained.

Source: https://assist.dla.mil -- Downloaded: 2026-10-26T07:55Z
Check the source to verify that this is the current version before use.

3. DEFINITIONS

3.1 Metallographic cross sections. Cross sections through the welds as shown in figures 1 through 8, which are cut perpendicular to the direction of welding, polished and etched as specified in ASTM E340. These samples show internal discontinuity, depth of weld penetration, weld size, and other characteristics of the weldment.

3.2 Partial-penetration joint design. The term "partial-penetration joint" as used in this standard is defined as a weld joint containing an intentionally unfused area. Examples of typical partial-penetration joint designs are shown in figure 9.

3.3 Penetrameter/Image Quality Indicator (IQI). A strip of metal the same composition as that of the metal being tested, representing a percentage of object thickness and provided with a combination of steps, holes, and/or slots. Its image on a radiograph is used to determine the radiographic quality level. It is not intended for use in judging the size nor for establishing acceptance limits of discontinuities. Examples of penetrameters/IQIs are shown in figures 2, 3, 4, 5 and 6.

3.4 Nondestructive testing terms and definitions. Nondestructive testing terms and definitions are as specified in ASTM E1316.

Source: https://assist.dla.mil -- Downloaded: 2026-10-26T07:55Z
Check the source to verify that this is the current version before use.

4. GENERAL REQUIREMENTS

4.1 Radiographic location. Weldments shall be radiographically inspected in the location specified on the applicable radiographic position chart drawing, component drawing, specification, or contract requirement. Radiographic inspection shall meet the requirements of MIL-STD-410 and ASTM E1742.

4.1.1 Radiographic frequency. Establishment of radiographic frequency of spot checking of weldments shall be accomplished in accordance with MIL-STD-1941. A Government approved quality assurance plan has commonly been used.

4.1.2 Angle beam. Radiographic inspection of partial-penetration welds requires special consideration of joint design when selecting the angle beam. The angle employed shall insure adequate coverage of the weld with minimal interference from the normally unfused area. Generally this can be achieved by directing the X-ray beam such that any possible incomplete penetration at both roots would be separated on the film by at least 0.125 in. (3.18 mm). Figure 1 illustrates the radiographic results of a correct and incorrect angle beam. If a radiographic position chart is not available, the recommended procedures shown on figure 9 shall be used as a guide.

4.1.3 Penetrameter/IQI requirements. Each penetrameter/IQI shall be produced with three holes, one of which shall be a diameter equal to twice the penetrameter/IQI thickness (2T). Penetrameters/IQI's shall conform to the requirements of ASTM E142 and ASTM E1742.

4.2 Acceptance standards. Designation of acceptance standards for each type of discontinuity illustrated in the reference standards shall be made by the design activity and approved by the Government procurement agency. The specific level of acceptance shall be designated by the "Standard Level" which is the minimum acceptance level for that discontinuity. The standard level shall be documented on the applicable radiographic position chart, drawing or contract requirement.

4.3 Discontinuity types. With the exception of cracks, the common discontinuities experienced in partial-penetration steel weldments are shown in the reference standards forming part of this document. These discontinuities are described below and in figures 2 through 6. All cracks shall be rejected; any deviation from this procedure will require authorization from the Government procuring activity.

a. Scattered porosity (fine and coarse). This discontinuity consists of scattered voids formed by gasses failing to escape during weld metal solidification. On the radiograph these discontinuities are dark round or elongated spots of varying size

Source: https://assist.dla.mil -- Downloaded: 2026-10-26T07:55Z
Check the source to verify that this is the current version before use.

and density. The average size of fine porosity is 0.031 in. (0.79 mm) in diameter and that of coarse porosity is 0.063 in. (1.59 mm) in diameter.

b. Clustered porosity. Clustered porosity appears the same as scattered porosity except that the pores are concentrated in one area, generally at the start of a bead or an interrupted arc.

c. Linear porosity. The type most common in partial-penetration welds results from expanding gasses generated in the normally unfused area. The cavities are linearly distributed at the root of the weld deposit and generally range from 0.031 in. (0.79 mm) in diameter to 0.063 in. (1.59 mm) in diameter.

d. Gas cavities. This discontinuity generally results from inadequate shielding gas or severe contamination of the gas. It appears the same as scattered porosity except that the average cavity size is approximately 0.125 in. (3.18 mm) in diameter.

e. Incomplete (inadequate) penetration. In partial penetration welds this discontinuity consists of a linear root void in excess of the unfused area normally present in this type of joint. It appears on the radiograph as a straight dark line at either root or at both roots. When present at both roots the two images should be separated if properly radiographed.

f. Lack of fusion. Lack of fusion is failure of the weld metal to fuse completely with the base metal or with the preceding bead. Lack of fusion resulting from trapped slag appears on the radiograph as a dark elongated line varying in width. Lack of fusion not accompanied by slag generally occurs along the weld interface and produces a hair-line unfused condition. This is difficult to reveal unless the plane of the discontinuity is parallel with the X-ray beam.

4.4 Discontinuity grading. Discontinuity severity grading of the reference standards was selected to extend from standard 1, which represents a high-quality weld, to that of standard 5, which is indicative of poor workmanship and is usually rejected in commercial practice.

Source: https://assist.dla.mil -- Downloaded: 2026-10-26T07:55Z
Check the source to verify that this is the current version before use.

5. DETAILED REQUIREMENTS

5.1 Application of reference standards.

5.1.1 Determination of acceptability. Acceptability of production welds shall be determined by directly comparing the production radiograph with the designated reference standard for each type of discontinuity. In general the extent of discontinuity exhibited in the designated standard will be permitted throughout the length of any particular weld joint provided that the discontinuity content in any weld length equal to that of the reference standard does not exceed that shown in the standard. Exceptions to this are required for incomplete penetration and lack of fusion since acceptability of those discontinuities is based on material thickness. The examples shown in figures 7 and 8 are for illustration purposes only; the evaluation is described in 5.1.3.

5.1.2 Quality level of radiographic sensitivity. Figures 2, 3, 4, 5 and 6 illustrate the radiographic image quality level of 2-1T. ASTM E1742 defines this formula as follows:

Radiographic quality level	2-1T
Penetrameter/IQI designation	1
Maximum penetrameter/IQI thickness expressed as a percentage of material thickness, T	2%
Minimum penetrameter/IQI hole diameter expressed as a multiple of thickness of penetrameter/IQI	1T
Equivalent penetrameter/IQI sensitivity expressed as a percentage of the specimen thickness in which a 2T hole would be clearly visible under the same radiographic conditions	1.4%

5.1.3 Rules 1, 2, and 3. Incomplete penetration and lack of fusion shall be judged by image width and length. Unless otherwise specified by the Government procuring activity the following rules shall apply:

Rule 1. The average image width of any incomplete penetration or lack of fusion shall not exceed 0.063 in. (1.59 mm).

Rule 2. All lines of incomplete penetration having an average of 0.032 in. (0.79 mm) or less shall be acceptable.

Source: https://assist.dla.mil -- Downloaded: 2026-10-26T07:55Z
Check the source to verify that this is the current version before use.

Rule 3. The greatest accumulated length of all incomplete penetration lines in any weld length of 8T, where T equals the average plate thickness, shall not exceed the following:

1T for standard 1
2T for standard 2
4T for standard 3
6T for standard 4
8T for standard 5

5.1.4 False indications. At times, radiographic films may indicate weld discontinuities when actually the film is defective. If doubt exists as to whether the discontinuity is in the weld or a film imperfection, then the weld shall be radiographed again.

5.2 Inspection records. Radiographic reports and films shall be made available to the procuring activity in accordance with ASTM E1742.

Source: https://assist.dla.mil -- Downloaded: 2026-10-26T07:55Z
Check the source to verify that this is the current version before use.

6. NOTES

(This section contains information of a general or explanatory nature which may be helpful, but is not mandatory.)

6.1 Intended use. This standard is intended to demonstrate various types and degrees of discontinuities occurring in partial-penetration steel welds and to give instructions concerning radiographic inspection of these types of weldments.

6.2 Subject term (key word) listing.

Arc welding
Discontinuity
Fusion Welding
Gas cavities
Inspection
Porosity
Angle Beams
Unfused Area
Weldments

6.3 Changes from previous issue. Marginal notations are not used in this revision to identify changes with respect to the previous issue due to the extent of the changes.

Source: https://assist.dla.mil -- Downloaded: 2026-10-26T07:55Z
Check the source to verify that this is the current version before use.

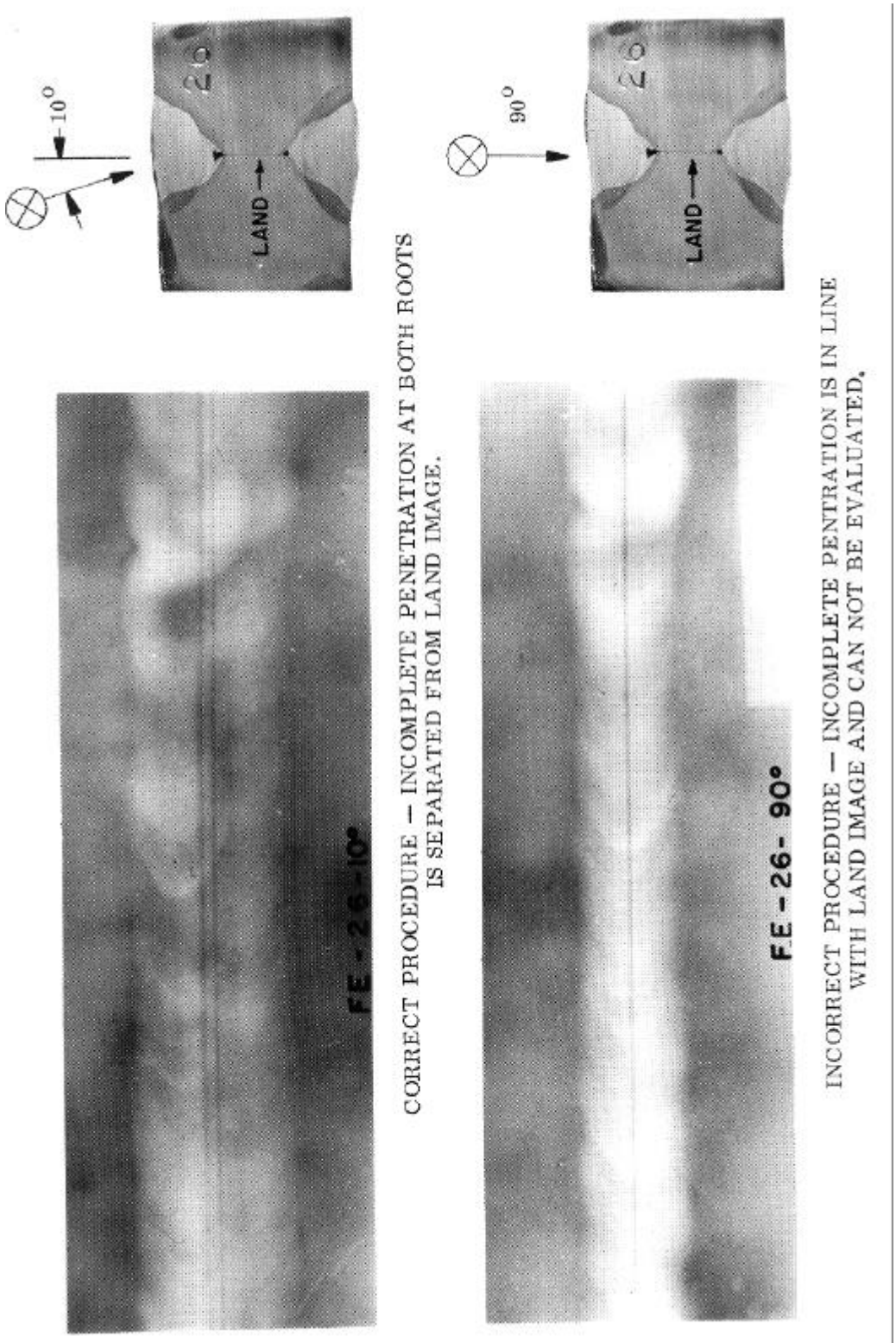
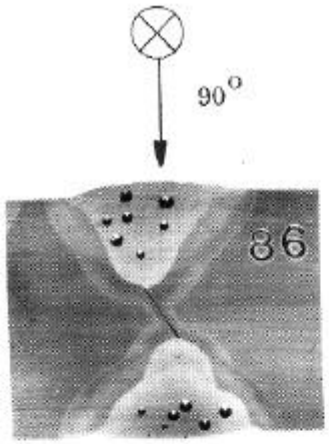
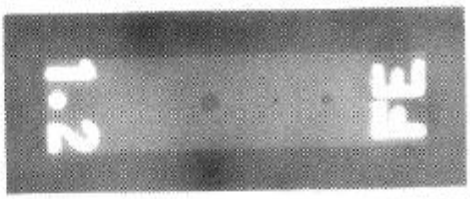


FIGURE 1. Correct and incorrect radiographic procedure for joint design (A).

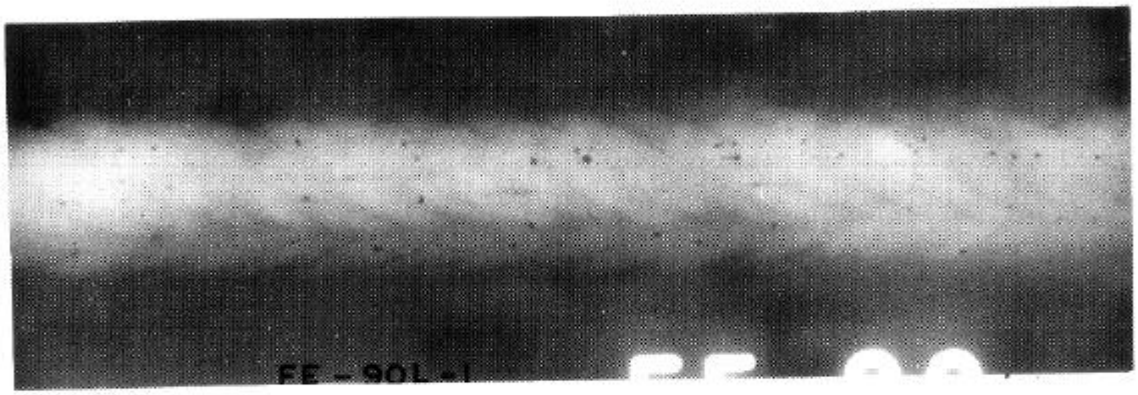
Source: https://assist.dla.mil -- Downloaded: 2026-10-26T07:55Z
Check the source to verify that this is the current version before use.



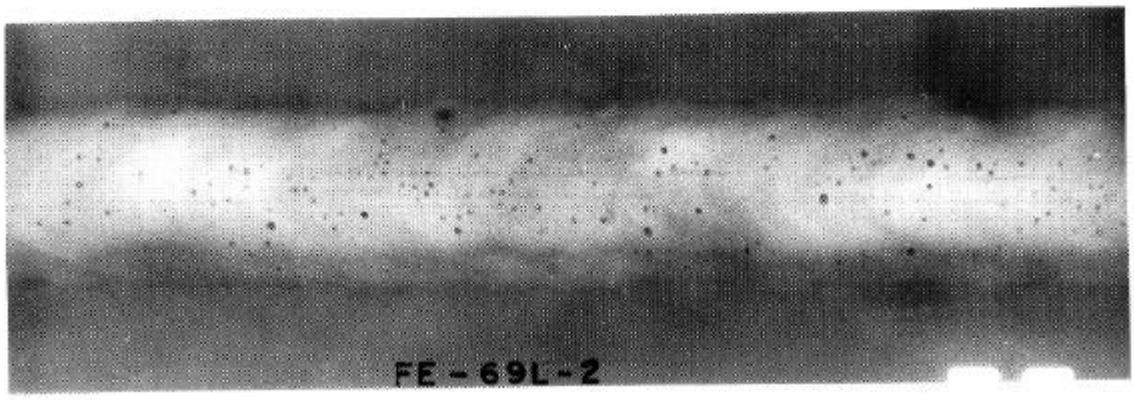


2-1T RADIOGRAPHIC SENSITIVITY

←CROSS SECTION OF FINE SCATTERED POROSITY. MOST PLATES ARE JOINT B, RADIOGRAPHED AS ILLUSTRATED.



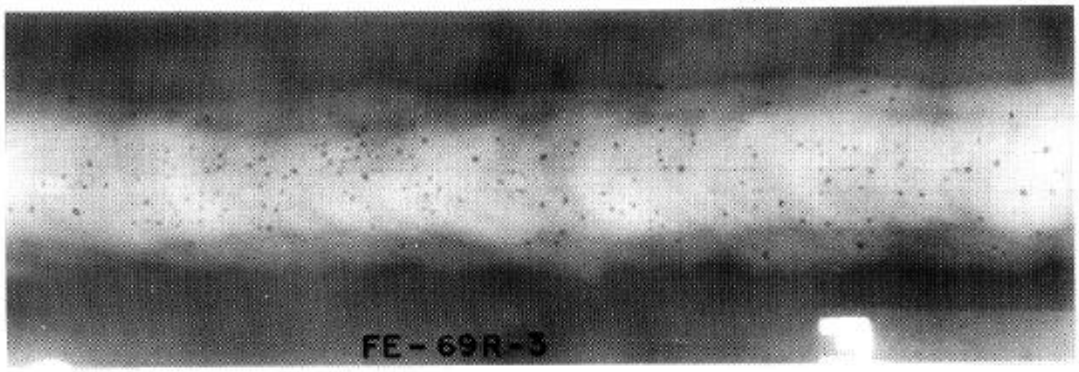
STD. 1 (APPROX. 6 PORES PER SQ. IN.)



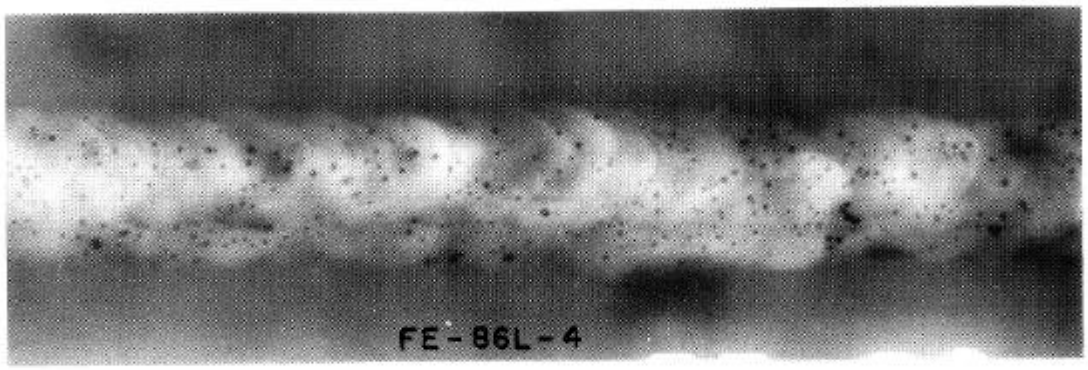
STD. 2 (APPROX. 12 PORES PER SQ. IN.)

FIGURE 2. Reference standards for fine scattered porosity in steel welds.

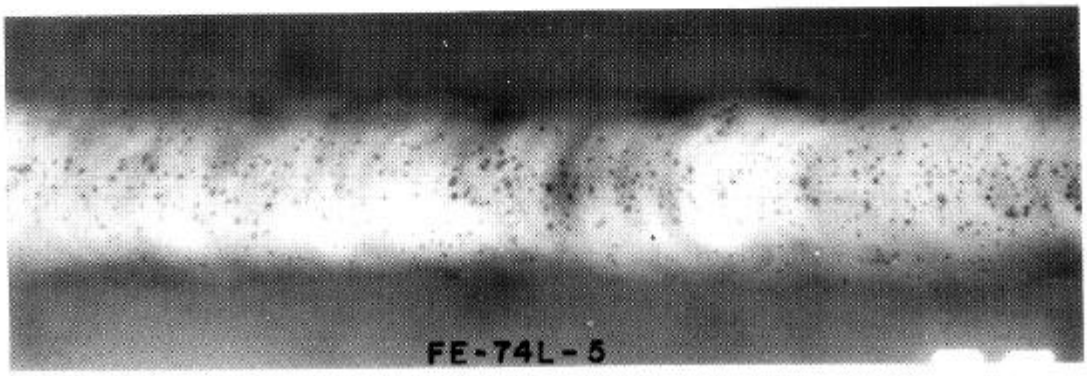
Source: https://assist.dla.mil -- Downloaded: 2026-10-26T07:55Z
Check the source to verify that this is the current version before use.



STD. 3 (APPROX. 25 PORES PER SQ. IN.)



STD. 4 (APPROX. 50 PORES PER SQ. IN.)



STD. 5 (APPROX. 100 PORES PER SQ. IN.)

FIGURE 2. Reference standards for fine scattered porosity in steel welds - Continued.

Source: https://assist.dla.mil -- Downloaded: 2026-10-26T07:55Z
Check the source to verify that this is the current version before use.

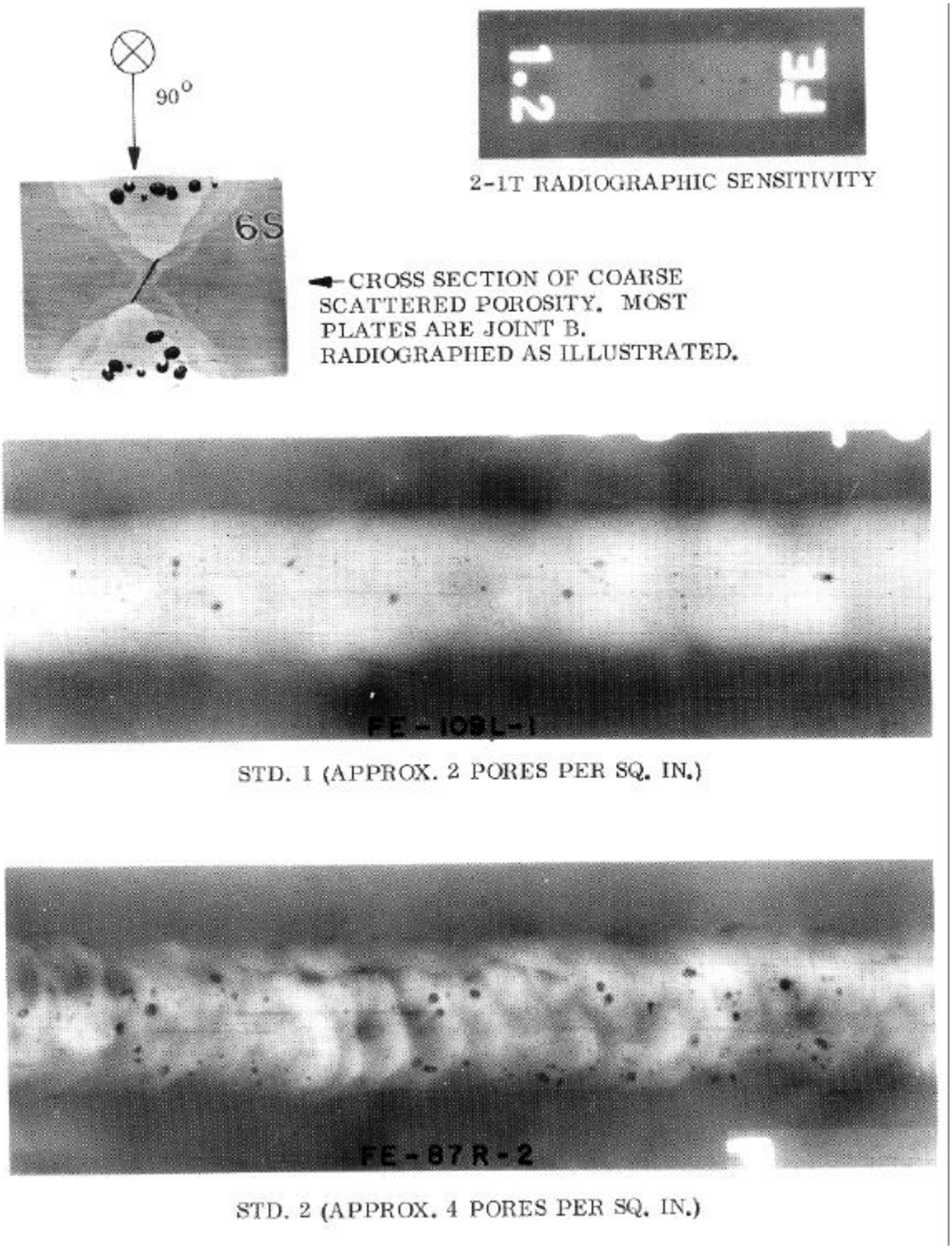
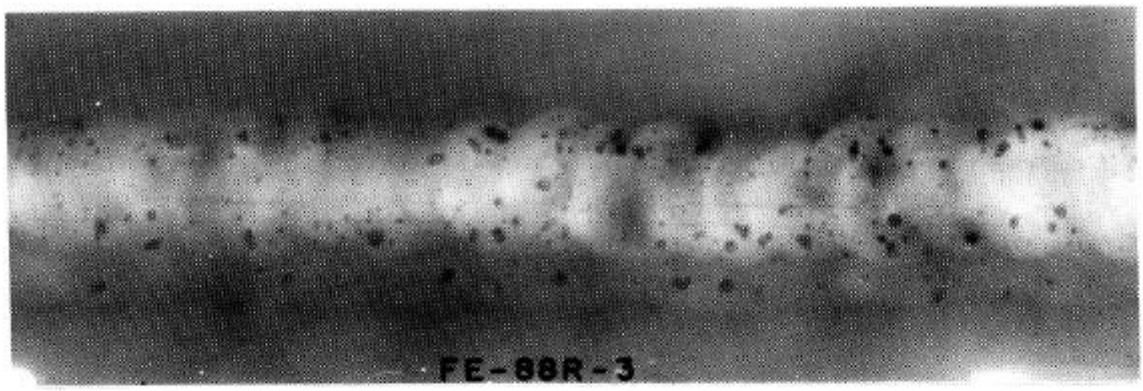
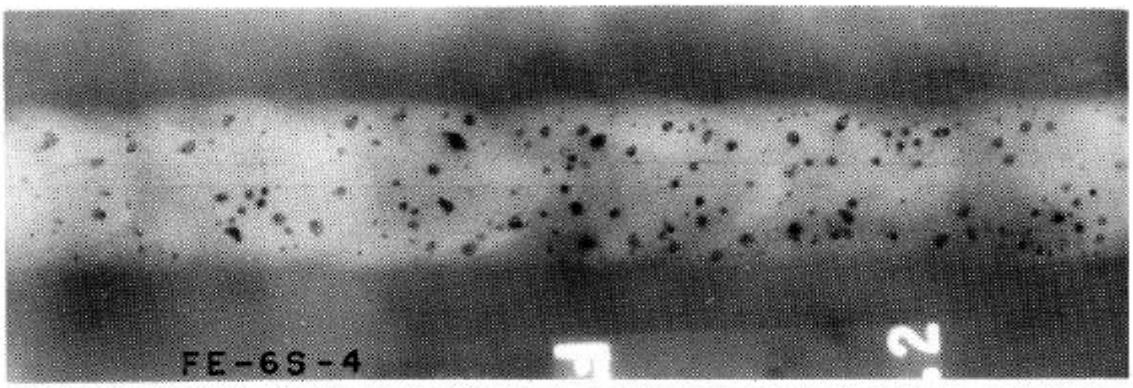


FIGURE 3. Reference standards for coarse scattered porosity in steel welds.

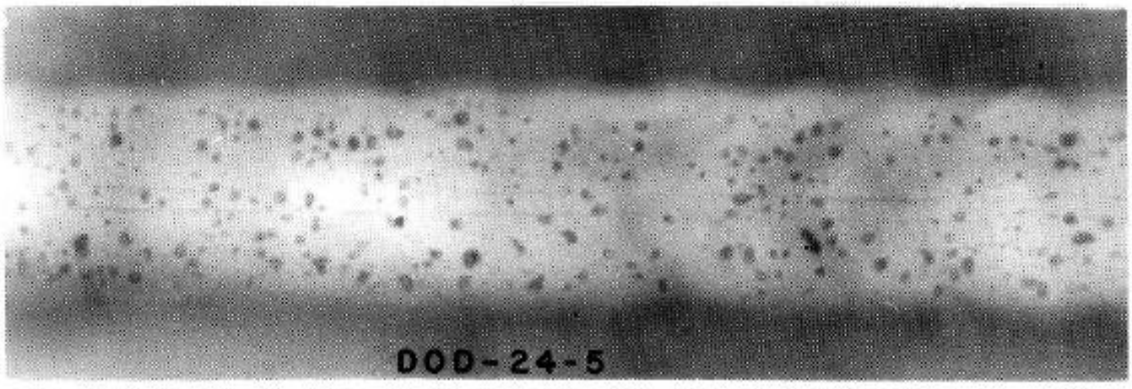
Source: https://assist.dla.mil -- Downloaded: 2026-10-26T07:55Z
Check the source to verify that this is the current version before use.



STD. 3 (APPROX. 8 PORES PER SQ. IN.)



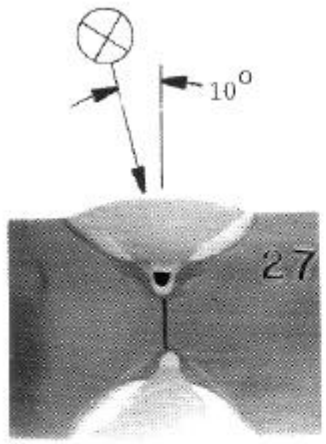
STD. 4 (APPROX. 16 PORES PER SQ. IN.)

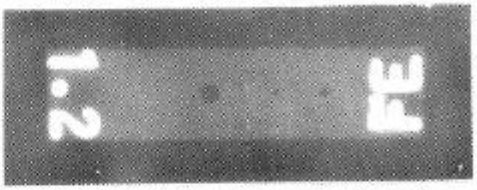


STD. 5 (APPROX. 32 PORES PER SQ. IN.)

FIGURE 3. Reference standards for coarse scattered porosity in steel welds - Continued.

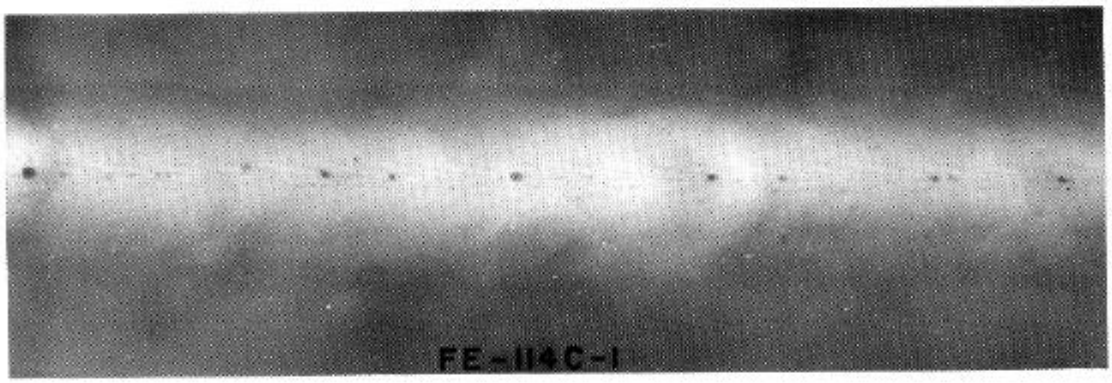
Source: https://assist.dla.mil -- Downloaded: 2026-10-26T07:55Z
Check the source to verify that this is the current version before use.



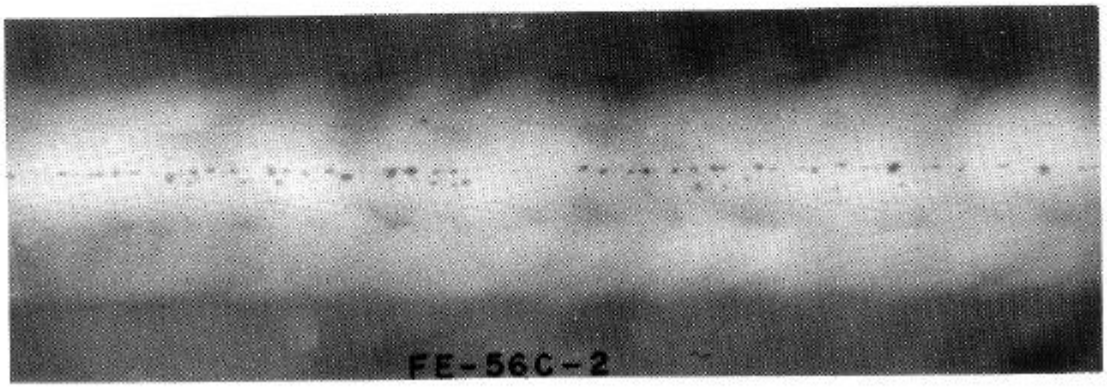


2-1T RADIOGRAPHIC SENSITIVITY

←CROSS SECTION OF LINEAR POROSITY. ALL PLATES ARE JOINT A, RADIOGRAPHED AS ILLUSTRATED.



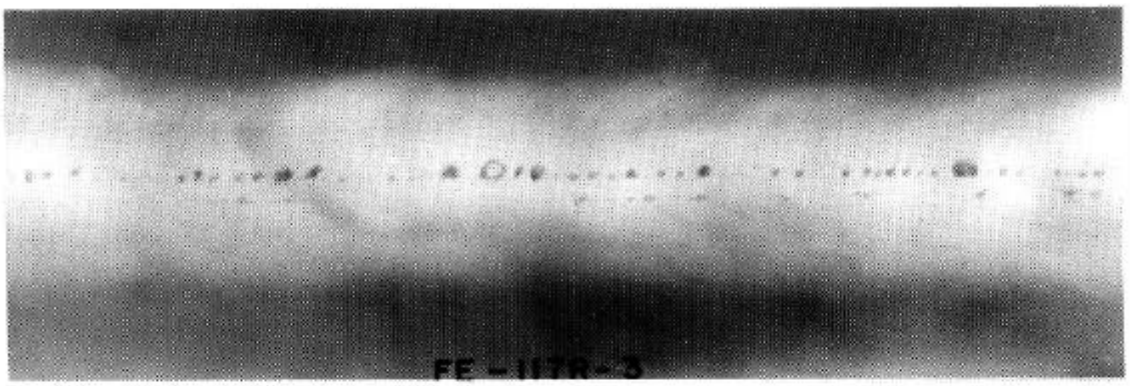
STD. 1



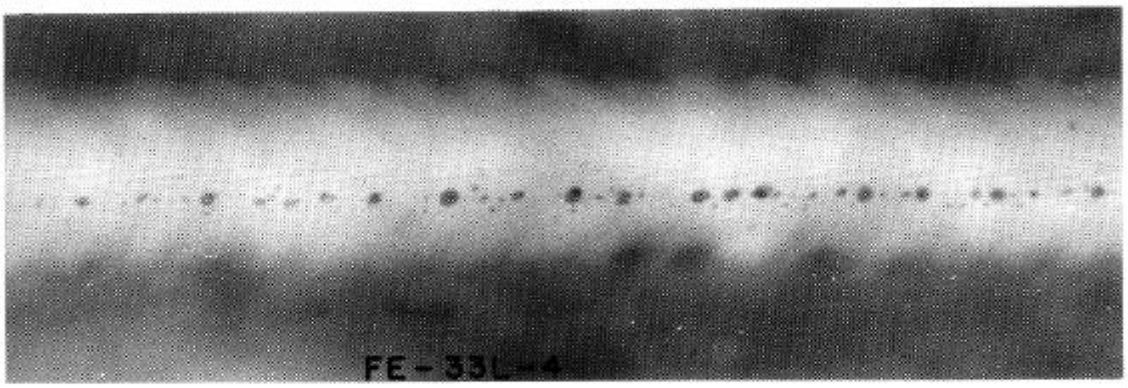
STD. 2

FIGURE 4. Reference standards for linear porosity in steel welds.

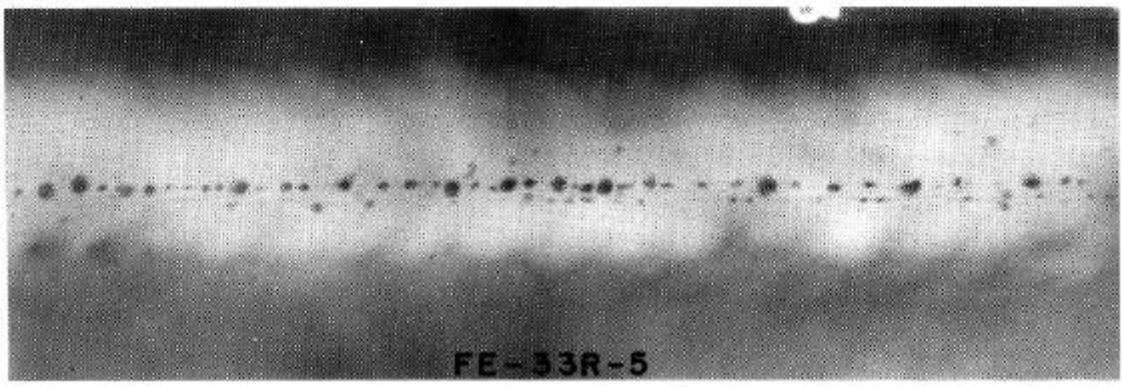
Source: https://assist.dla.mil -- Downloaded: 2026-10-26T07:55Z
Check the source to verify that this is the current version before use.



STD. 3



STD. 4



STD. 5

FIGURE 4. Reference standards for linear porosity in steel welds - Continued.

Source: https://assist.dla.mil -- Downloaded: 2026-10-26T07:55Z
Check the source to verify that this is the current version before use.

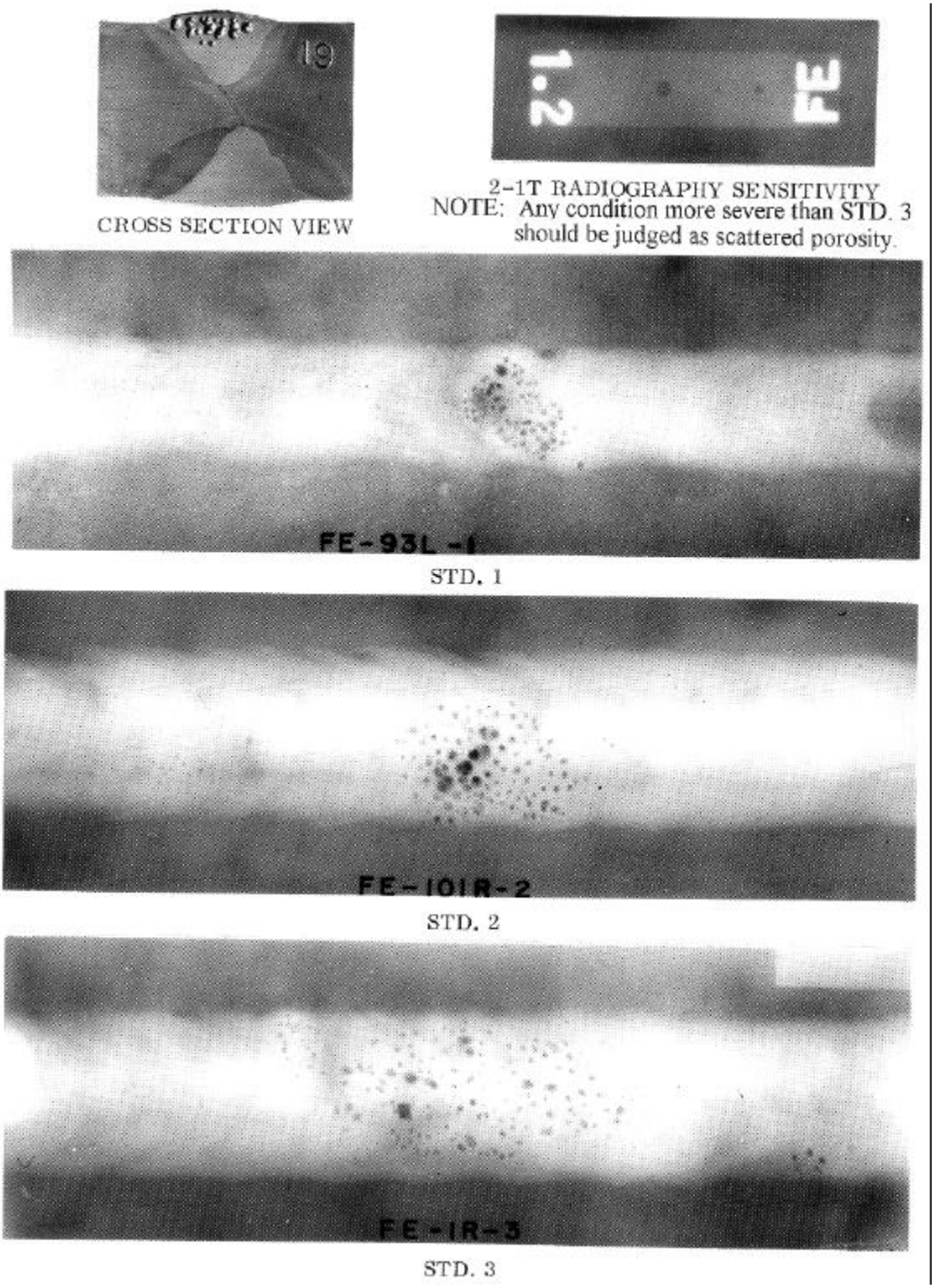


FIGURE 5. Reference standards for clustered porosity in steel welds.

Source: https://assist.dla.mil -- Downloaded: 2026-10-26T07:55Z
Check the source to verify that this is the current version before use.

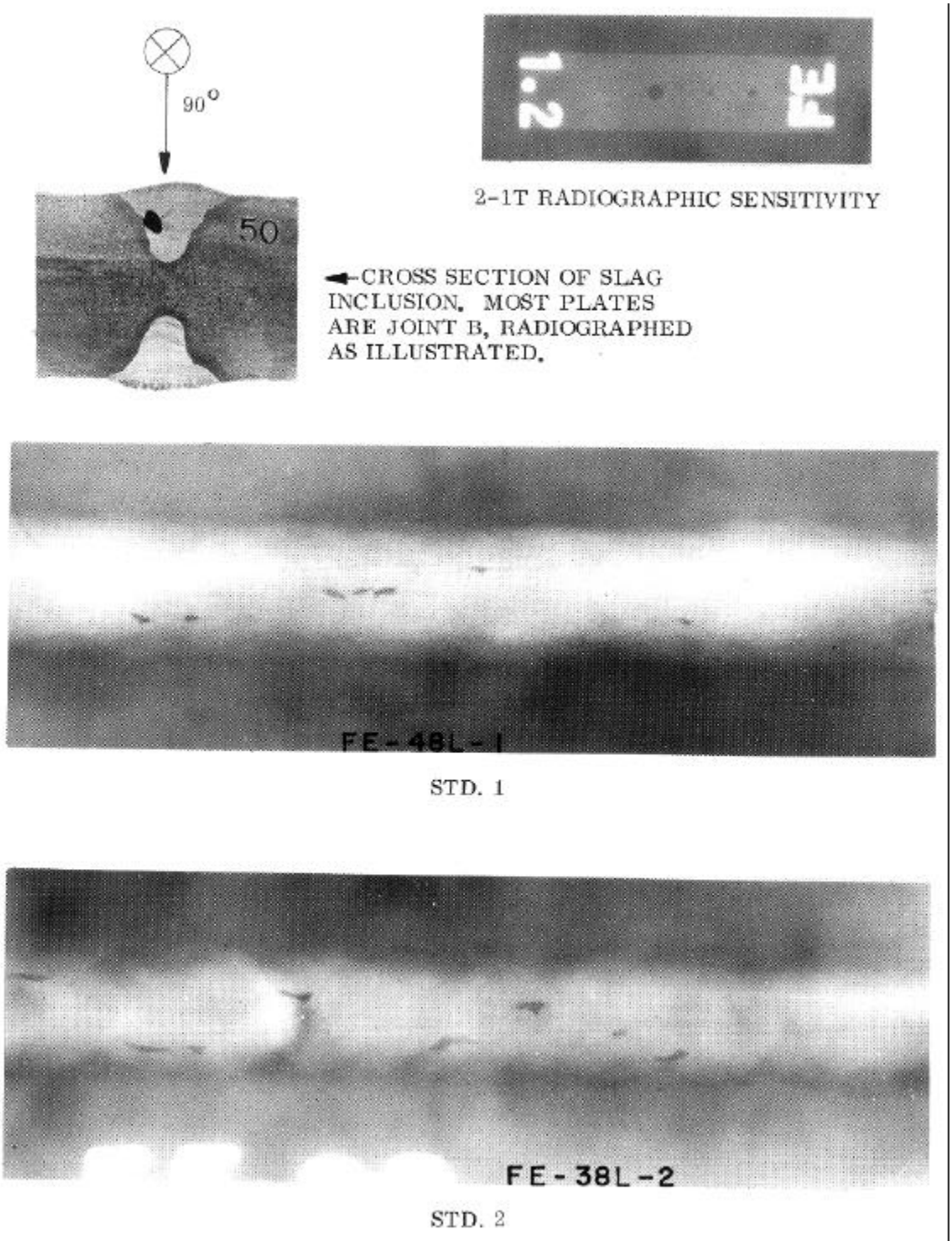
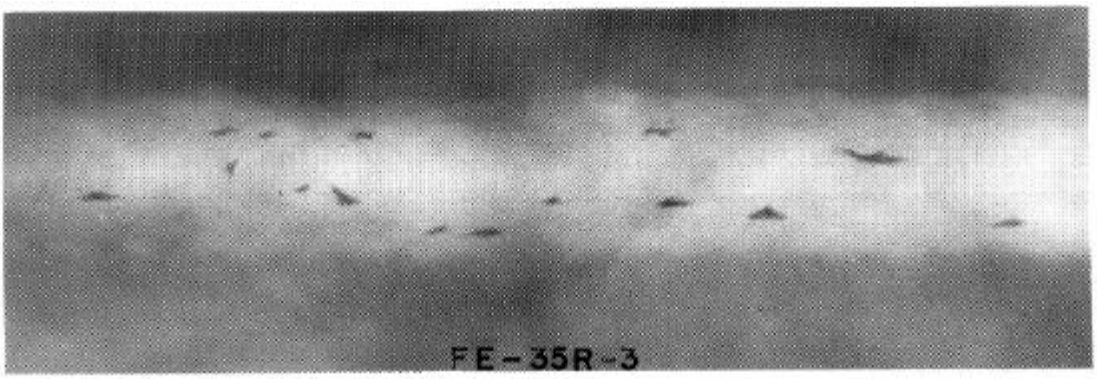
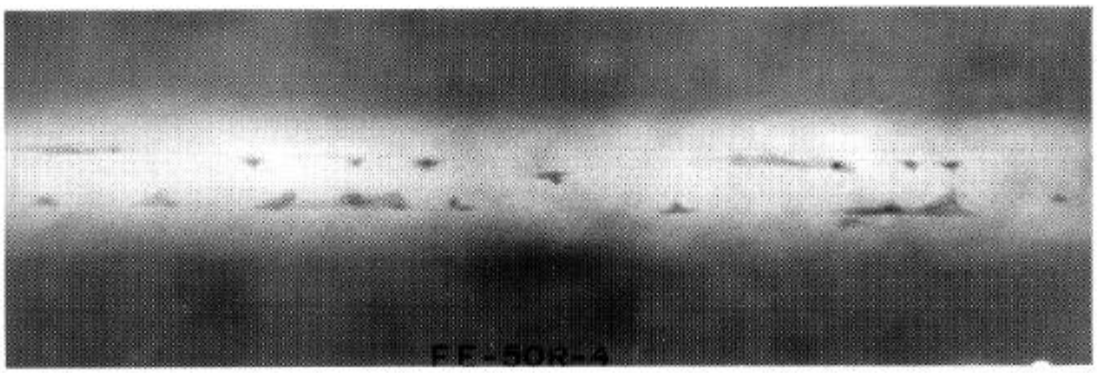


FIGURE 6. Reference standards for scattered slag inclusions in steel welds.

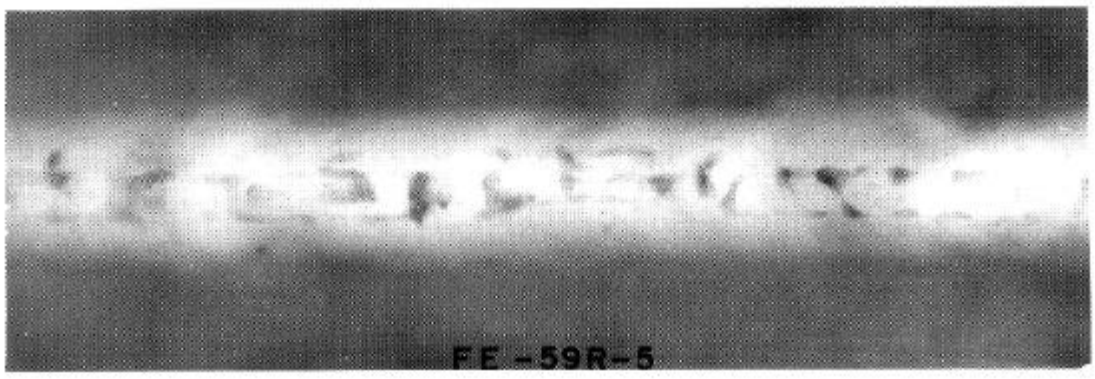
Source: https://assist.dla.mil -- Downloaded: 2026-10-26T07:55Z
Check the source to verify that this is the current version before use.



STD. 3



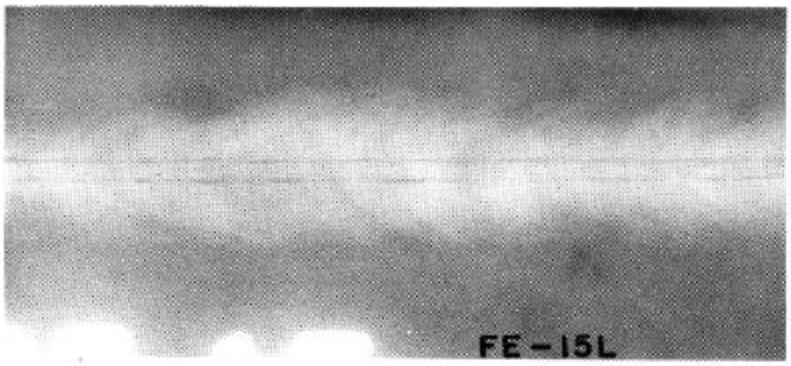
STD. 4



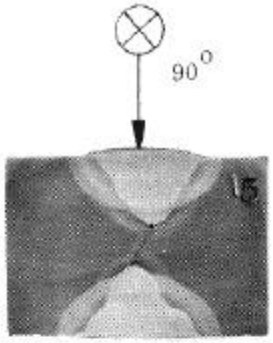
STD. 5

FIGURE 6. Reference standards for scattered slag inclusion in steel welds - Continued.

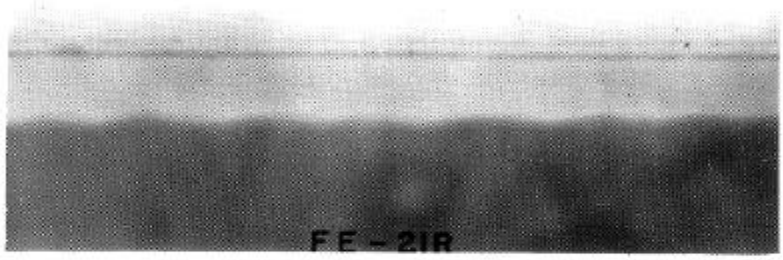
Source: https://assist.dla.mil -- Downloaded: 2026-10-26T07:55Z
Check the source to verify that this is the current version before use.



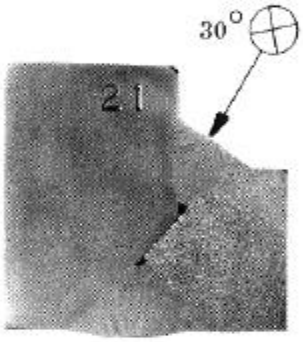
SLIGHT CONDITION — NORMAL FOR THIS JOINT USING STICK ELECTRODE.



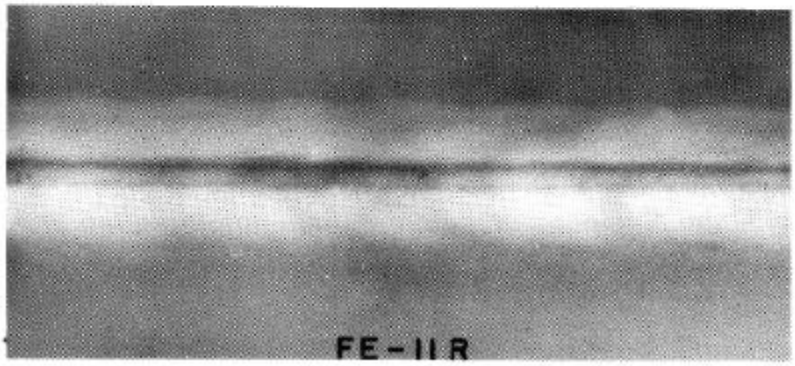
JOINT B



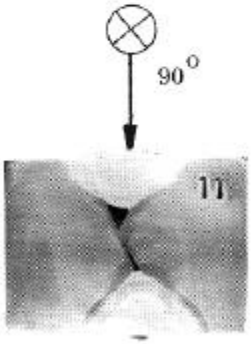
MODERATE CONDITION — RESULTING FROM TOO LARGE AN ELECTRODE ON ROOT PASSES.



JOINT L



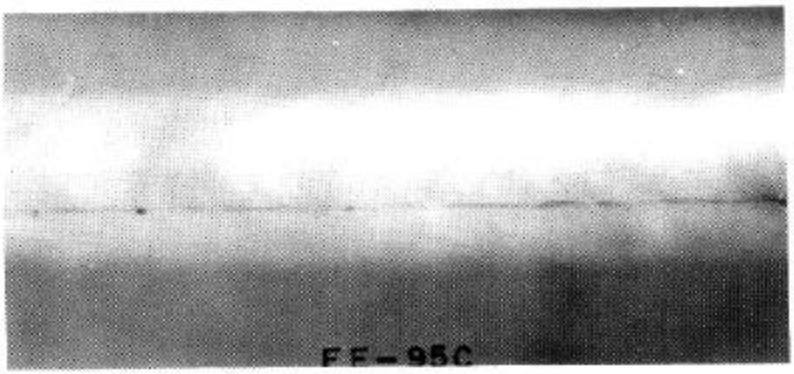
GROSS CONDITION — RESULTING FROM LOW HEAT AND TOO LARGE AN ELECTRODE.



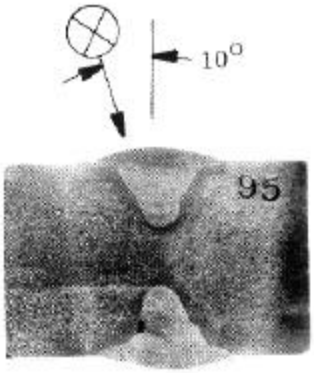
See 5.1.3 (Rules 1, 2 & 3) for severity ratings.

FIGURE 7. Varying degrees of incomplete penetration in steel welds correlated with weld cross section.

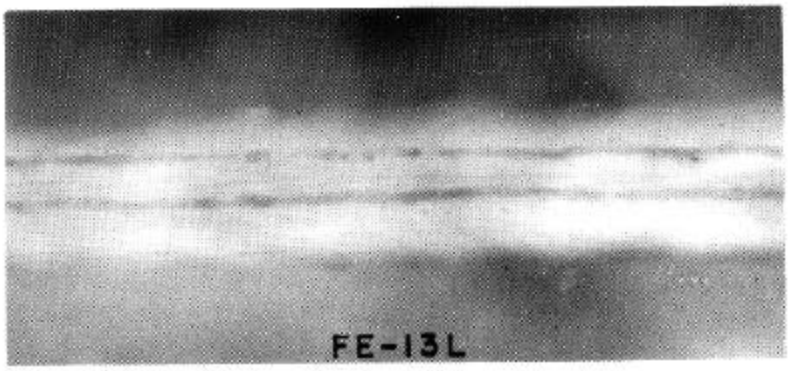
Source: https://assist.dla.mil -- Downloaded: 2026-10-26T07:55Z
Check the source to verify that this is the current version before use.



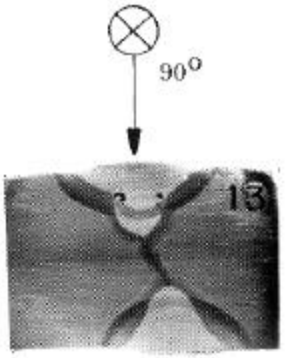
LACK OF FUSION CAUSED BY TRAPPED SLAG BETWEEN WELD BEADS.



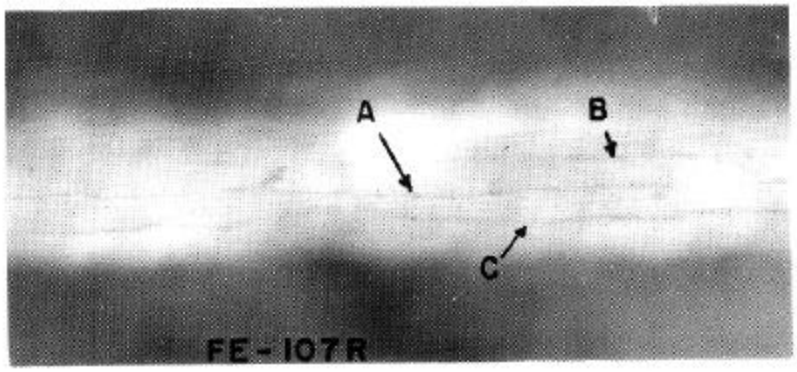
JOINT A



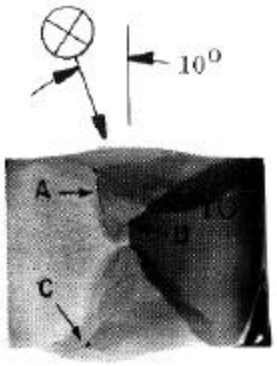
SLAG TRAPPED IN SHARP NOTCHES PRODUCED BY A HIGH CROWN BEAD.



JOINT B



SEVERE CONDITION DUE TO POOR GUN ANGLE, ALSO SHARP NOTCH EFFECT AT C.



JOINT A

See 5.1.3 (Rules 1, 2 & 3) for severity ratings.

FIGURE 8. Lack of fusion in steel welds correlated with weld cross sections.

Source: https://assist.dla.mil -- Downloaded: 2026-10-26T07:55Z
Check the source to verify that this is the current version before use.

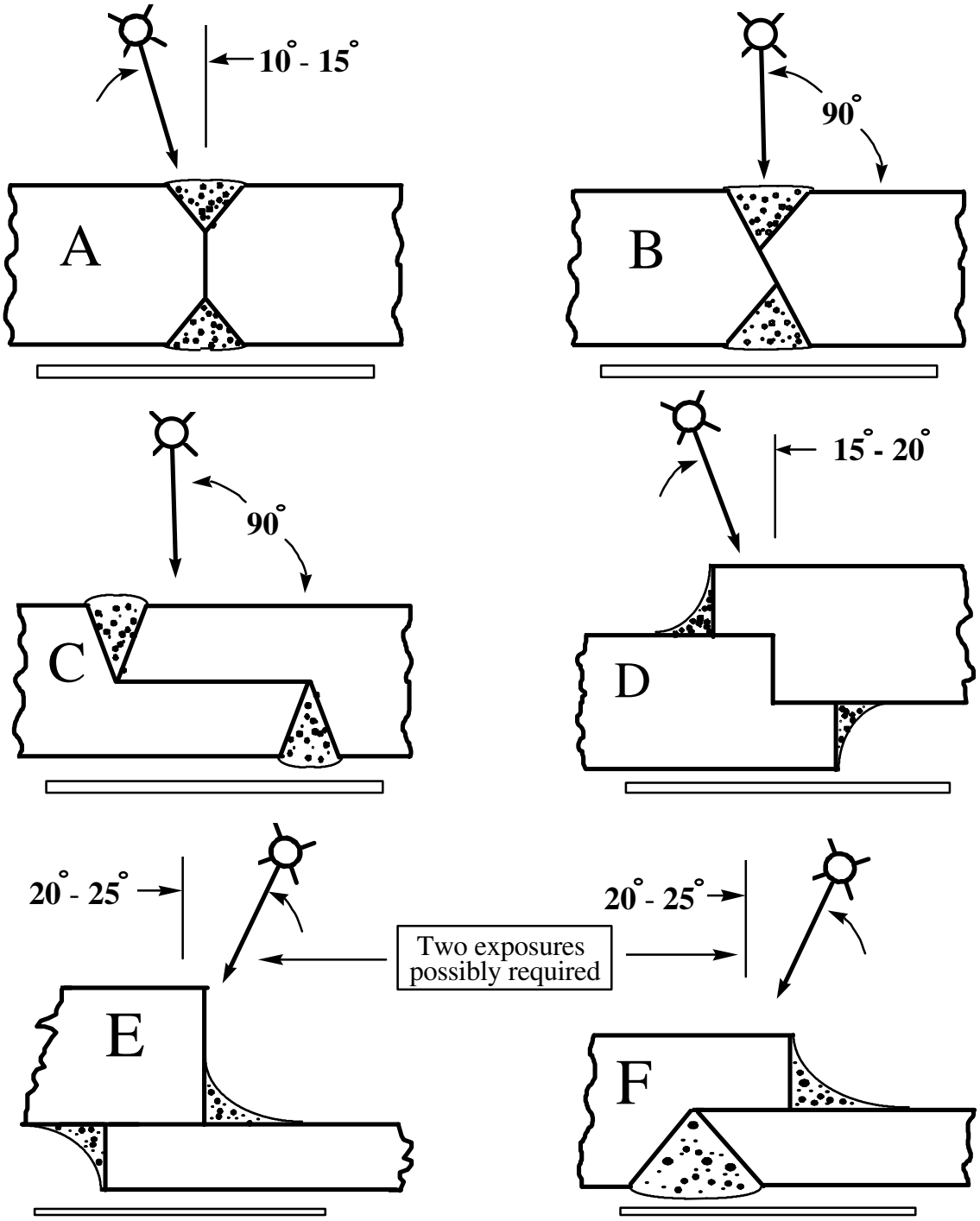


FIGURE 9. Recommended angle beam and film location for typical partial-penetration joints.

Source: https://assist.dla.mil -- Downloaded: 2026-10-26T07:55Z
Check the source to verify that this is the current version before use.

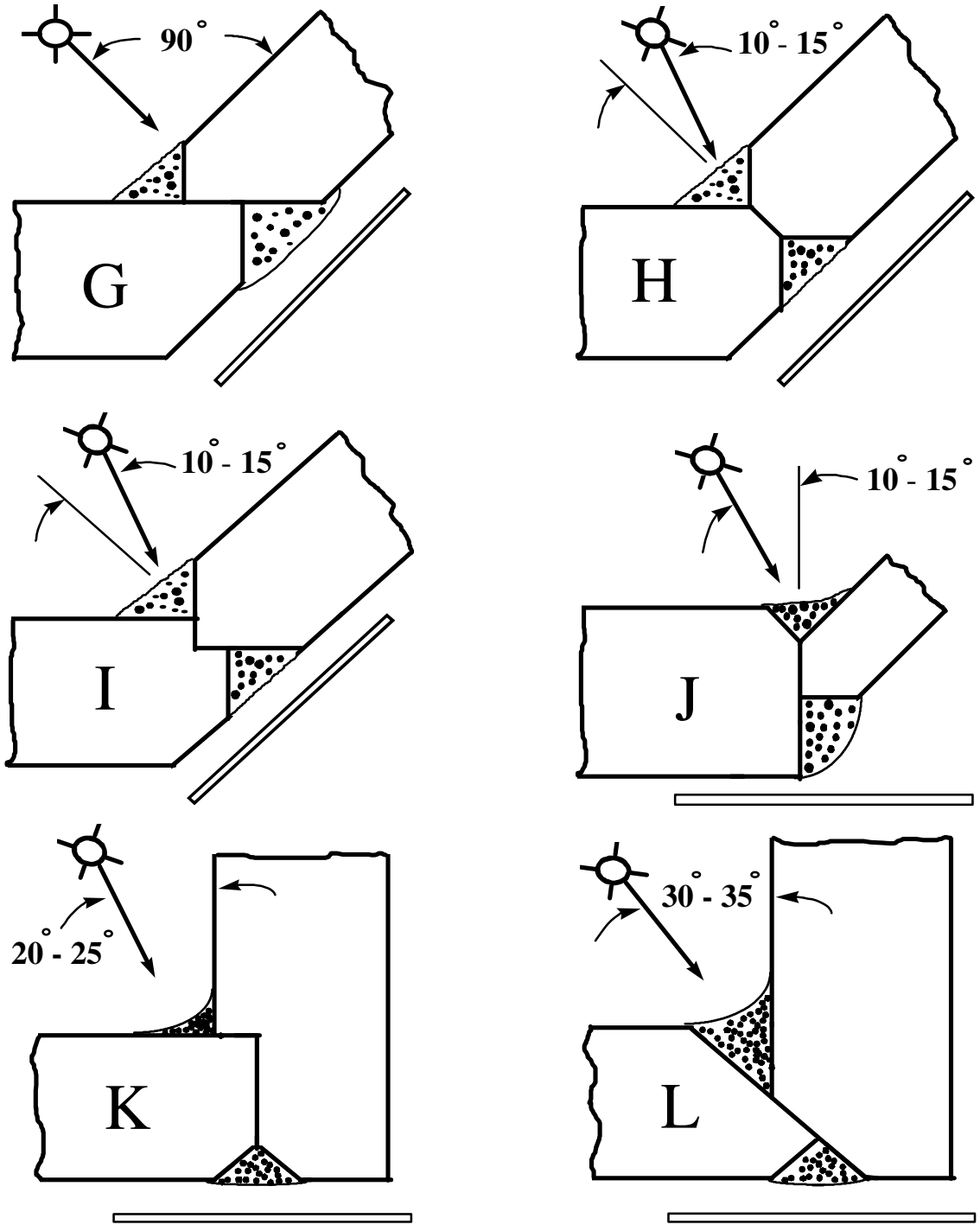


FIGURE 9. Recommended angle beam and film location for typical partial-penetration joints - Continued.

Source: https://assist.dla.mil -- Downloaded: 2026-10-26T07:55Z
Check the source to verify that this is the current version before use.

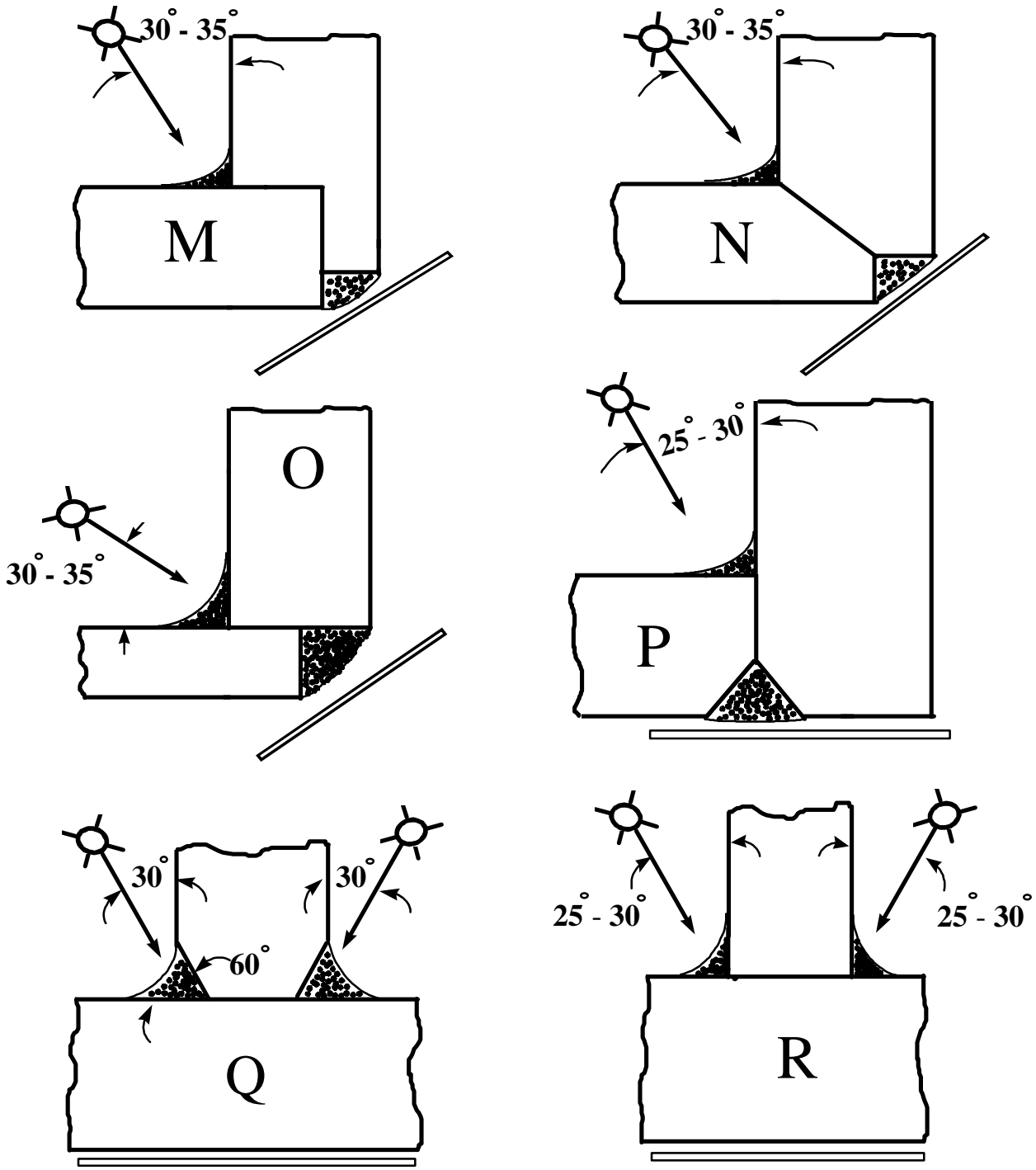


FIGURE 9. Recommended angle beam and film location for typical partial-penetration joints - Continued.

Source: https://assist.dla.mil -- Downloaded: 2026-10-26T07:55Z
Check the source to verify that this is the current version before use.

Custodian:
Army - AT
Air Force - 11

Review Activities:
Army - AR

Preparing Activity:
Army - AT

(Project NDTI-0264)

Source: https://assist.dla.mil -- Downloaded: 2026-10-26T07:55Z
Check the source to verify that this is the current version before use.

STANDARDIZATION DOCUMENT IMPROVEMENT PROPOSAL

INSTRUCTIONS

1. The preparing activity must complete blocks 1, 2, 3, and 8. In block 1, both the document number and revision letter should be given.
2. The submitter of this form must complete blocks 4, 5, and 7.
3. The preparing activity must provide a reply within 30 days from receipt of the form.

NOTE: This form may not be used to request copies of documents, nor to request waivers, or clarification of requirements on current contracts. Comments submitted on this form do not constitute or imply authorization to waive any portion of the referenced document(s) or to amend contractual requirements.

I RECOMMEND A CHANGE:	**1. DOCUMENT NUMBER** MIL-STD-1894B	**2. DOCUMENT DATE *(YYMMDD)*** 980626

3. DOCUMENT TITLE
RADIOGRAPHIC REFERENCE STANDARDS AND RADIOGRAPHIC PROCEDURES FOR STEEL WELDS

4. NATURE OF CHANGE ***(Identify paragraph number and include proposed rewrite, if possible. Attach extra sheets as needed.)***

5. REASON FOR RECOMMENDATION

6. SUBMITTER

a. NAME *(Last, First, Middle Initial)*	**b. ORGANIZATION**	
c. ADDRESS *(Include Zip Code)*	**d. TELEPHONE *(Include Area Code)*** **(1) Commercial** **(2) AUTOVON** ***(If applicable)***	**7. DATE SUBMITTED *(YYMMDD)***

8. PREPARING ACTIVITY

a. NAME	**b. TELEPHONE *(Include Area Code)*** **(1) Commercial** (810) 574-8745 **(2) AUTOVON** 786-8745
c. ADDRESS *(Include Zip Code)* **Commander** **U.S. Army Tank-automotive and Armaments Command** **ATTN: AMSTA-TR-E/BLUE** **Warren, MI 48397-5000**	**IF YOU DO NOT RECEIVE A REPLY WITHIN 45 DAYS, CONTACT:** **Defense Quality and Standardization Office** **5203 Leesburg Pike, Suite 1403** **Falls Church, VA 22041-3466** **Telephone (703) 756-2340 AUTOVON 289-2340**

Source: https://assist.dla.mil -- Downloaded: 2026-10-26T07:55Z
Check the source to verify that this is the current version before use.